Project Guidelines

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Your goal is to make an R package that can be used for some practical task by a user. As part of your submission, you will submit a `.tar.gz` file that should compile/load without any issue along with a 1 minute video reel demonstrating the package.

The marks distribution (20 marks):

- 10 marks for the uniquess of the idea, creativity, correctness, and implementation of the package
- 5 marks for the readability of the help files and documentations
- 5 marks for the creativity and information in the video reel.

Due Date: April 4th, 5:00pm

You will learn how to make an R package in a worksheet separately. But essentially, an R package is just a bunch of functions that can be loaded in memory and used by the user. More information on the package and video to be submitted will be given later.

Type of package

You are free to use your creativity as to what purpose the package will serve. There are certain restrictions:

- The central technique employed must have been taught/discussed in this course or MTH210. This can include
  - PCA, UMAP, tSNE, k-means, EM algorithm, kNN, simulation, CLT, consistency, unbiasedness, sampling, cross-validation, gradient-descent
- You may add a Shiny Dashboard if appropriate
- You may deal with images, sound files, videos, etc in the package, if appropriate

Example package

Here is my **example** package idea:

`GetColorPalette`: You can upload any image into the package and it will return the color palette of the image. This can be extremely useful for shops like IKEA to display the color themes in front of a model room. This is a SIMPLE package to be honest, and is just for demonstration. You may think of other creative ideas!

```
library(GetColorPalette)
```

Loading the package above, we can now call the function `get_palette` that I have in this package. One can also do `?get_palette` to open the help page:

```r
library(imager)

living_room <- load.image("best-IKEA-living-room.jpg")
plot(living_room)
```


```r
# getting the palette with 5 colors
get_palette(living_room, k = 5)
```

```r
# getting the palette with 10 colors
get_palette(living_room, k = 10)
```